ESP32-C3 · SOP8 · WEBSERIAL

# Your ESP32. Three builds.

Choose a build for your SuperMini. All variants use the same SOP8 connector, with a power jumper and 10 kΩ CS pull-up.

**Unverified build candidate**
Rev. 0.4 · Hardware validation pending

A

## A · Base adapter

Perfboard base with a removable ESP32-C3.

→

B

## B · Top-mounted adapter

Compact adapter above the SuperMini.

→

C

## C · EPLZON

EPLZON board with isolated copper groups.

→

VARIANT A · ASSEMBLY · USB ON TOP

# ESP32-C3 → SOP8
# Minimal perfboard adapter

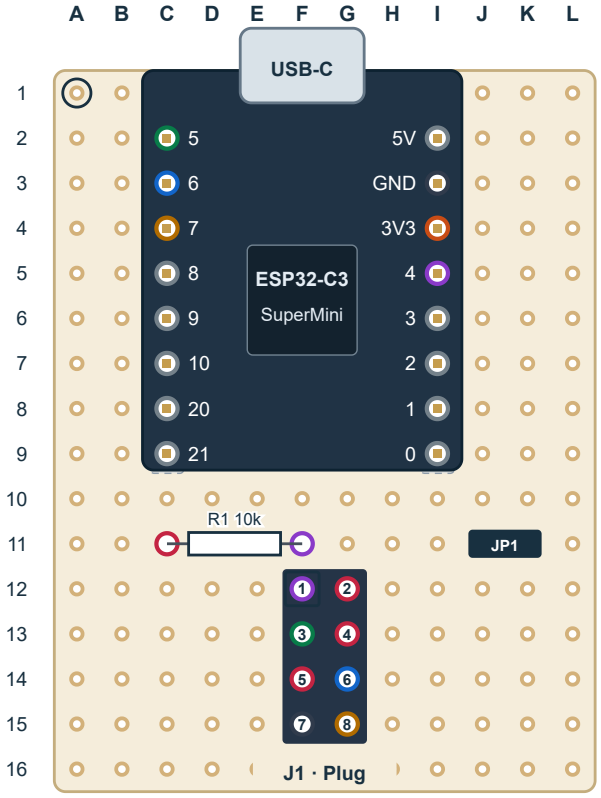
For your **ESP32-C3 SuperMini** and the existing SOP8 cable. The adapter is inserted into **TB2 Dungeon Overworld** as ESP32-C3 used.

> **Rev. 0.4 replaces Rev. 0.1–0.3:** CS-Pull-up added and the swapped ESP32 pin rows corrected. PCB component side up, USB up: **5V/GND/3V3 right**, GPIO5/6/7 on the left. Do not use previous hole coordinates.

Unverified build candidate · Hardware test open

## POPULATING · USB EDGE TOP

Place components on this side. Wires go to the opposite solder side.




30,48 mm · 12 Columns

40.64 mm · 16 rows · 2.54 mm pitch

### Each hole is individually isolated.

### SuperMini · plug-in

Jack strips: C2–C9 and I2–I9
USB-C up; 5V remains unconnected.

### J1 · 8 contacts

F12/F13/F14/F15 = 1/3/5/7
G12/G13/G14/G15 = 2/4/6/8

### Power Jumper JP1

J11 = 3V3 from the module; K11 = to the target
Fit the jumper cap to power the target.

### Cable orientation

Cable contact 1 on marked pin 1.
Measure up to the clip.

### Mechanical fit test

Marked: Row spacing 15,24 mm.
Align the sockets on the module before soldering.

A1-Circle: same physical corner in both views.

Rev. 0.4 · Unverified Build Candidate · Only for the documented 3.3-V SOP8 wiring

Component mounting view: Module and connector are on top. The sockets lie under the module. Pin 1 on the connector is marked with a permanent dot.

**PCB selection:** Variants A and B require separate, unconnected pads. The EPLZON board shown, with connected groups of five, uses Alternative C. Before populating, check copper groups.

# Parts list and CS-Pull-up

| Quantity | Part | Execution |
|---|---|---|
| 1 | Perfboard | Individual copper pads; 2.54 mm pitch; 12 × 16 holes; approx. 30.5 × 40.6 mm |
| 2 | Jack strip | Straight, 1×8, 2.54 mm; compatible with module pins |
| 1 | Connector J1 | Straight, 2×4, 2.54 mm; eight open contacts; compatible with the existing cable |
| 1 | Power Jumper JP1 | 1×2 pin header, 2.54 mm, plus matching jumper |
| 1 | R1 · CS-Pull-up | 10 kΩ, axial, ≥ 1/8 W, 1–5 %; body max. 6,5 × 2,5 mm; C11/F11 |
| 1 | Switching wire | Approx. 0.5 m, insulated, approx. 0.14 mm² / AWG26; eleven short pieces |
| 1 | ESP32-C3 SuperMini | Available; if not populated, additionally two straight pin strips 1×8 |
| 1 | SOP8 cable with clamp | Available; wiring as in the table, measure through |

Tool: Soldering iron, solder, side cutters, wire stripper, and multimeter. If needed, cut and sand the board before populating. Around 15 minutes is a target for prepared parts, no time guarantee for the first build.

## As with the flipper: keep CS on HIGH

The flash uses the same signals as in the flipper setup. R1 connects CS through **10 kΩ to VCC**, also the power behind the jumper. The GPIO numbers of the ESP32 are different; the function on the chip remains the same.




The resistor is an additional branch. **The CS line is not interrupted by a series resistor.** W10 and W11 connect the two resistor legs to the corresponding connector pins. R1 is located above between C11 and F11 and has no polarity.

## Check the view before soldering

Component side of the SuperMini is up, USB is up: pins on the left start with GPIO5/6/7; on the right with 5V/GND/3V3. The solder side drawing mirrors only left/right. It does not change hole labels or connector pin numbers.

> With the ESP32, cables and target disconnected, measure between connector pins 1 and 2: expect 9.5–10.5 kΩ with a 5% R1. With the jumper open, the module's 3V3 remains disconnected from the target supply. Physical hardware validation is still pending.

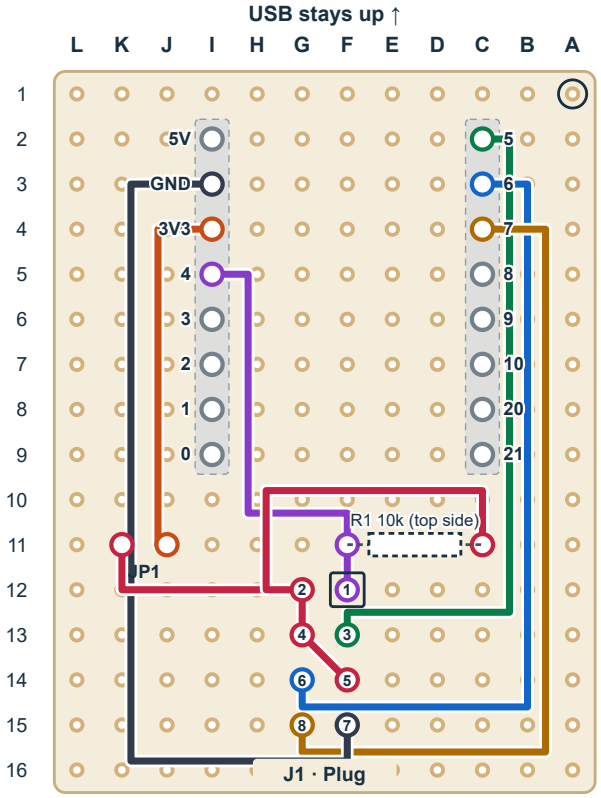
VARIANT A · SOLDER SIDE · USB REMAINS ON TOP

# Eleven insulated wire pieces and R1

The board **turn left/right**, so that the USB edge remains on top. The columns now run from L to A. Hole **C5 remains C5**, although it is now on the right.

## SOLDER SIDE · USB EDGE TOP

Turn left/right, not top/bottom. Coordinates remain on the same hole.




30,48 mm · 12 Columns

40.64 mm · 16 rows · 2.54 mm pitch

### Each hole is individually isolated.

Only solder the marked end points.

Lines show insulated wires, no traces.

- W1 I5 → F12 CS
- W2 C2 → F13 MISO
- W3 C3 → G14 CLK
- W4 C4 → G15 MOSI
- W5 I3 → F15 GND
- W6 I4 → J11 3V3
- W7 K11 → G12 VTARGET
- W8 G12 → G13 VTARGET
- W9 G13 → F14 VTARGET
- W10 F12 → F11 CS
- W11 G12 → C11 VTARGET

Crossings without solder point remain separated.

Use insulated wire for W8 and W9 as well.

The module is mounted on the other side.

Unused module pins remain unconnected.

Rev. 0.4 · Unverified Build Candidate · Only for the documented 3.3-V SOP8 wiring

Mirrored view of the solder side. The lines represent suggested routes for insulated wires. Solder only the endpoints; crossings and copper pads passed over by a wire must stay unconnected.

| Wire | Signal | From | Hole | After | Hole |
|---|---|---|---|---|---|
| W1 | CS | SuperMini GPIO4 | I5 | J1.1 | F12 |
| W2 | MISO | SuperMini GPIO5 | C2 | J1.3 | F13 |
| W3 | CLK | SuperMini GPIO6 | C3 | J1.6 | G14 |
| W4 | MOSI | SuperMini GPIO7 | C4 | J1.8 | G15 |
| W5 | GND | SuperMini GND | I3 | J1.7 | F15 |
| W6 | 3V3 | SuperMini 3V3 | I4 | JP1.1 | J11 |
| W7 | VTARGET | JP1.2 | K11 | J1.2 | G12 |
| W8 | VTARGET | J1.2 | G12 | J1.4 | G13 |
| W9 | VTARGET | J1.4 | G13 | J1.5 | F14 |
| W10 | CS | J1.1 | F12 | R1.CS | F11 |
| W11 | VTARGET | J1.2 | G12 | R1.VTARGET | C11 |

**JP1 remains unconnected below.** The jumper connects the two pins only on the populated side. W8 and W9 route the power behind JP1 to HOLD and WP. These short connections should also be made separately.

VARIANT A · ASSEMBLE & CHECK

# Short soldering instructions

1. **Prepare without power.** Remove SuperMini-USB, take off SOP8 cable, remove jumper. Disconnect battery and USB from the box.
2. **Passprobe.** Both sockets are plugged into the module and are set together in C2–C9 and I2–I9. USB is facing upwards. The drawing assumes a 15,24 mm row spacing and the shown pin layout. The module must not be compressed. Do not solder according to incorrect hole coordinates if the module variant differs.
3. **Attach the sockets.** Solder each outer contact briefly, check the fit, pull the module off without power, and solder the remaining contacts. If the module lacks pin headers, align them straight and solder them first.
4. **Insert connectors and JP1.** Connectors on F12–G15, JP1 on J11/K11. Both on the component side; solder from below. Pin 1 on F12 and the physical corner A1 mark. The cable must later sit flush over all eight contacts.
5. **R1 einsetzen.** 10 kΩ between C11 and F11, body top, solder pads bottom. No polarity. W10 connects F11 to CS; W11 connects C11 to the target power behind JP1. Do not add a 10-kΩ pull-up if one already exists in the cable.
6. **Solder wires W1–W11.** Strip only the short wire ends, so no bare wire touches adjacent solder pads. Keep the wires flat and short, with insulation intact at crossings. Solder all wire ends shown sharing a point to the same pad; keep the wire insulated outside each joint.
7. **Visual inspection and continuity.** Leave the module, cables and jumper disconnected at first. Make sure there are no solder bridges between adjacent pads and no wire scraps. Then take the following measurements.

**R1 is an additional branch:** CS remains directly connected to GPIO4 and chip pin 1. R1 pulls CS to HIGH (chip not selected) when GPIO is in high impedance; never place R1 in series on the CS wire. The other side belongs to target VCC, not before the power jumper.

## Connector pins and chip pins are numbered differently

| SuperMini / Power | Connector pin | SOP8-Chippin | Signal |
|---|---|---|---|
| GPIO4 | 1 | 1 | CS |
| GPIO5 | 3 | 2 | MISO |
| GPIO6 | 6 | 6 | CLK |
| GPIO7 | 8 | 5 | MOSI |
| GND | 7 | 4 | GND |
| 3V3 behind JP1 | 2 | 8 | VCC |
| 3V3 behind JP1 | 5 | 3 | WP |
| 3V3 behind JP1 | 4 | 7 | HOLD |

## Without power, module removed

□ W1–W11 have a connection between their endpoints.

□ Connectors 2/4/5 are connected. CS (Connector 1) to Connector 2: **9,5–10,5 kΩ** at 5-%-R1; no short. Other signal nets and GND remain separated. Measure without module, cable and target.

□ JP1 open: Module 3V3 is disconnected from pins 2/4/5. JP1 plugged in: connection exists.

□ 5V and all unused module pins remain unconnected.

□ Set up cables, leave the clip free: measure each row of the table until the correct clip contact. Color stripe alone is not enough for assignment.

## Check power supply

□ Insert the module without power. Plug in JP1 and connect the SuperMini-USB: about 3.3 V should be present between pin 2 (+) and 7 (GND).

□ Unplug the USB before attaching. The Box-Akku and Box-USB remain separate. Align pin 1 of the clip with the pin-1 marking on the chip.

□ First connect the SuperMini-USB and measure the voltage at the target under load. If there is a drop, heating, or module restarts, disconnect and check the power supply.

**An open power jumper is not a complete separation.** SPI and GND remain connected; backfeeding through SPI is possible. Unplug USB before rewiring. The load capacity of this specific SuperMini regulator has not yet been tested with the target. Here WP and HOLD are tied to 3.3 V for the documented target wiring; this is not general wiring guidance for arbitrary powered devices.

ALTERNATIVE B · MOUNTING ABOVE THE ESP32

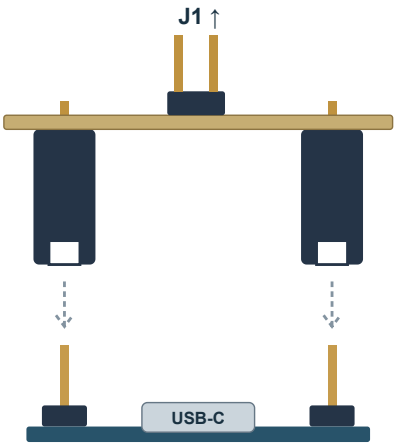
# High sockets on the cover, normal pins on the SuperMini

The SuperMini stays at the bottom. The small adapter is plugged in from above. **The high sockets belong under the cover**; their openings face downward. On the ESP32, the normal long contact pins face upward.

Alternative B · Unverified Build Candidate

## B · SIDE EXPLOSION VIEW

Schematically separated. Jumper stays on the header; normal pins stay on the ESP32.




View from the USB edge; two row of sockets

**Attachment: Connector + Jumper + R1 on top**

Mini board approx. 23 × 28 mm

**High sockets under the cover**

Case ≥ 8,5 mm; openings downward

To separate, pull off the entire insert.

**Normal ESP32 pins point upwards**

Long ends plug into the sockets

**The SuperMini stays underneath; USB remains accessible.**

**Check at least 2 mm distance between the mount solder pads/wires and ESP32 components.**

Disconnect USB before plugging in or unplugging. For BOOT/RESET, remove the clip if needed. Not to scale.

The exploded view shows the assignment of the parts. Jumpers and R1 are behind the connector and are not shown in this cross-sectional view. Check fit and spacing on the existing module.

## Parts list for Alternative B

| Quantity | Part | Execution |
|---|---|---|
| 1 | Attachment board | Individual plated-through solder pads; 2.54 mm pitch; 9 × 11 holes; approx. 22.86 × 27.94 mm |
| 2 | High socket strip | 1×8, 2.54 mm; housing height at least 8.5 mm; under the cover, openings facing down |
| 2 | Normal header strip | 1×8, 2.54 mm; into the SuperMini, long contact ends up; existing matching pins usable |
| 1 | Connector J1 | 2×4 open contacts, 2,54 mm; on top of the header |
| 1 | Power Jumper JP1 | 1×2 pin header plus jumper, 2.54 mm; top at the rear edge |
| 1 | R1 · CS-Pull-up | 10 kΩ, axial, ≥ 1/8 W, 1–5 %; body max. 6,5 × 2,5 mm; B10/E10, top |
| 1 | Insulated switch wire | Eleven pieces; approx. 0.5 m stock; thin enough for free holes, e.g. AWG30 with suitable insulation |
| 1 | SuperMini and SOP8 cable | Available; same electrical layout as variant A |

**Individual pads, no connected groups of five.** The copper surfaces of different holes must not already be connected. A board in a breadboard pattern with five connected holes per group is unsuitable for this layout. The top and bottom sides of the same hole must instead be through-contacted.

Grid: 2.54 mm. Hole coordinates and pin layout apply to the SuperMini with 15.24 mm spacing between rows of contacts. The adapter measures nominally 22.86 × 27.94 mm; cut edges outside the outer solder pads are left as-is. Temporarily insert the cable connector over all eight connector pins and check for free access to JP1.

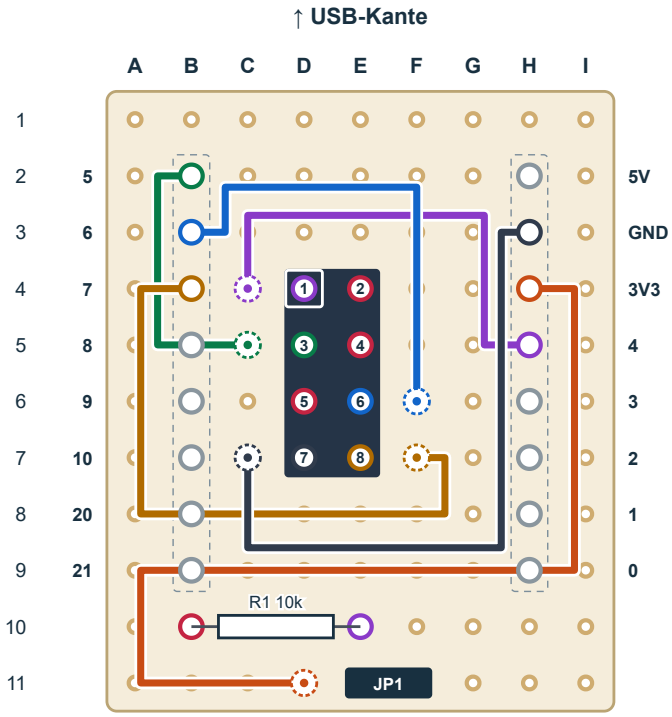
ALTERNATIVE B · TOP SIDE & WIRING

# Solder top, lead down isolated

The solder connections of the two high sockets come through the board from below and are **top** soldered. W1–W6 start there. The connector and JP1 are on top; their solder ends are on the bottom.

## B · TOP ATTACHMENT · USB ON TOP

High sockets under the board; their connections are soldered here on top.




22.86 × 27.94 mm · 9 × 11 holes

2.54 mm pitch · individual plated-through pads

### Switch sides without additional soldering point

Struck-through circle = free through-hole.

Run the wire through there, do not solder.

- **W1 H5 → C4 ↓ → D4**
- **W2 B2 → C5 ↓ → D5**
- **W3 B3 → F6 ↓ → E6**
- **W4 B4 → F7 ↓ → E7**
- **W5 H3 → C7 ↓ → D7**
- **W6 H4 → D11 ↓ → E11**
- **W7 F11 → E4 underside only**
- **W8 E4 → E5 underside only**
- **W9 E5 → D6 only bottom**
- **W10 D4 → E10 underside only**
- **W11 E4 → B10 only bottom**

W1–W6: solder from the through-hole to the target pin.

J1 and JP1 are on this top side.

Mark pin 1; set the cable over all eight pins.

Crossings of isolated wires remain separated.

Alternative B · Rev. 0.4 · Unverified Build Candidate · Pinout like Variant A

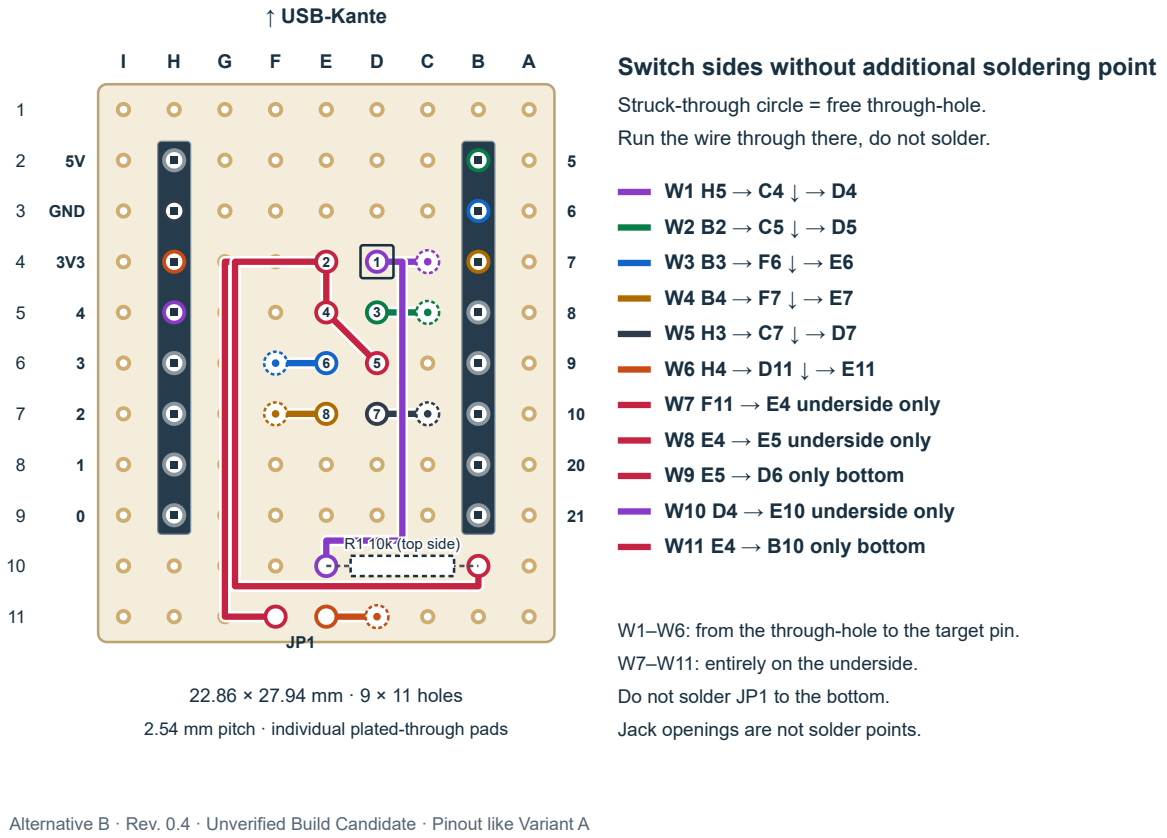
Pass-through holes are unused holes for the untinned, fully insulated section of a wire. There are eleven wire segments plus R1; no extra solder joint is made at a pass-through hole.

| Wire | Signal | From | Hole | Pass-through hole ↓ | After | Hole |
|---|---|---|---|---|---|---|
| W1 | CS | ESP GPIO4 | H5 | C4 | J1.1 | D4 |
| W2 | MISO | ESP GPIO5 | B2 | C5 | J1.3 | D5 |
| W3 | CLK | ESP GPIO6 | B3 | F6 | J1.6 | E6 |
| W4 | MOSI | ESP GPIO7 | B4 | F7 | J1.8 | E7 |
| W5 | GND | ESP GND | H3 | C7 | J1.7 | D7 |
| W6 | 3V3 | ESP 3V3 | H4 | D11 | JP1.1 | E11 |
| W7 | VTARGET | JP1.2 | F11 | – | J1.2 | E4 |
| W8 | VTARGET | J1.2 | E4 | – | J1.4 | E5 |
| W9 | VTARGET | J1.4 | E5 | – | J1.5 | D6 |
| W10 | CS | J1.1 | D4 | – | R1.CS | E10 |
| W11 | VTARGET | J1.2 | E4 | – | R1.VTARGET | B10 |

**No connection change:** Connector 1 → CS/GPIO4; 3 → MISO/GPIO5; 6 → CLK/GPIO6; 8 → MOSI/GPIO7; 7 → GND. Connector 2/4/5 receive 3.3 V behind JP1. The SOP8 chip assignment is in the common table in section 3.

ALTERNATIVE B · BOTTOM SIDE & ASSEMBLY

# Flip the attachment, leave the USB edge on top




The underside is mirrored left/right. The black rows are the downward-facing sockets. Their solder connections are on the top side.

## Soldering sequence for Alternative B

1. **Align without power.** Insert normal pins into the SuperMini: short solder ends through its board, long contact ends upwards. Align straight and solder from below. Existing matching pins can remain.
2. **Attach the sockets on the adapter.** High sockets from below on B2–B9 and H2–H9, openings downward. Align with the power-off ESP32, solder one end contact on top briefly. Remove the cover, then solder the remaining connections on top.
3. **Mount the connector and JP1.** Connector on top at D4–E7, jumper on top at E11/F11 set and solder from below. Pin 1 on D4 and the USB edge permanently mark.
4. **R1 montieren.** 10 kΩ, body top between B10 and E10; solder bottom. W10: D4 → E10 (CS). W11: E4 → B10 (target power behind JP1). No polarity; do not duplicate an existing 10-kΩ cable pull-up.
5. **W1–W11 connect.** W1–W6 solder at the top to the connector solder pad, guide through the marked free hole with undamaged insulation, lead down to the target pin and solder there. W7–W11 lie entirely on the bottom. Use thin wires; do not force through a too narrow hole.
6. **Check and mount.** Without module, cables, and target D4 → E4 measure on 9.5–10.5 kΩ. Direct CS wire remains low-ohmic, open JP1 disconnects Modul-3V3. Remaining continuity and short-circuit test according to the common test in section 3 and B-coordinates. Only then plug into the power-off SuperMini. Check at least 2 mm distance of all header solder points and wires to ESP32 components; no bare wire loops downward.

> **WebSerial applies to all three variants.** Then perform the shared firmware, voltage and read checks in sections 3 and 4. Disconnect USB before attaching or removing the module. An open jumper only disconnects power to the target; it does not prevent backfeeding through SPI.

**Open:** real module fitting, position of the pin contacts, cable/jumper freedom, solder points, power under load and real read comparisons. Achieve BOOT/RESET by pulling off the power-off module if needed. 15 minutes is a target for prepared parts, no guaranteed first-time setup time.

ALTERNATIVE C · EPL2ON AS CARRIER BOARD

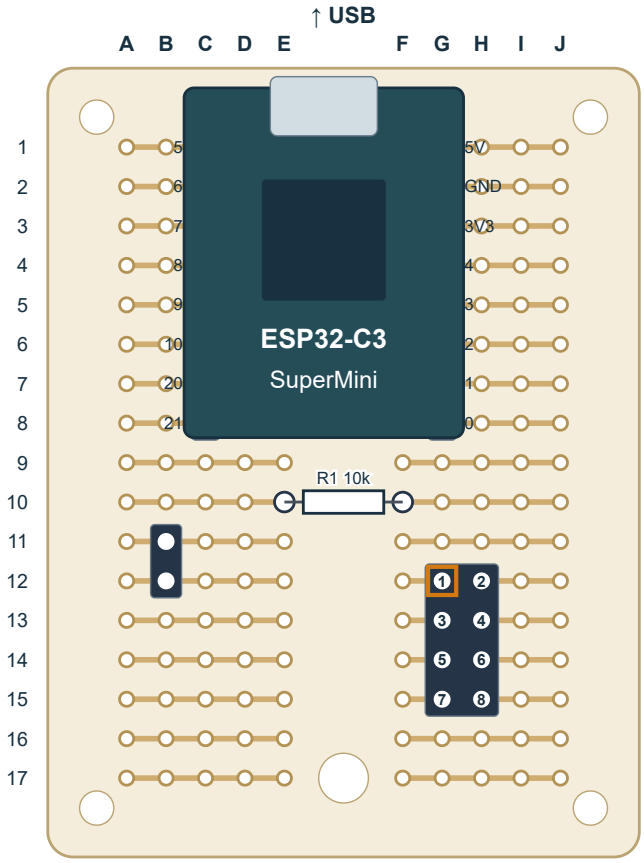
# Insert SuperMini on top

The whole **38.1 × 50.8 mm board** remain in place. Two sockets hold the ESP32-C3. The plug and power jumper are next to them on the same top side. All eleven wires are below.

Alternative C · Rev. 0.4 · Unverified Build Candidate

## C · ESP32 MOUNTED

All coordinates retain their designation from the top side. USB edge remains on top.




38.1 × 50.8 mm · 2.54 mm pitch

### Plug-in on the carrier board

ESP32: pins facing down.

Jacks: top, openings facing up.

USB points to the upper board edge.

### Power-Jumper left bottom

B11: Input from the 3,3-V pin.

B12: Output to the target.

### Connector at bottom right

Watch for open pins in the 2×4 grid.

Orange marking: Pin 1 at G12.

Set the cable over all eight pins.

Pin 1 wire to the marked side.

Schematic view of the module.

Check fit and USB access.

Alternative C · Rev. 0.4 · Unverified Build Candidate · Components on top, wires on bottom

Sockets on top, long ESP32 pins down. USB edge and Pin 1 permanently mark. The Pin-1 wire of the tested cable must go to G12; align the connector separately on Chip pin 1.

## Parts list for Alternative C

| Quantity | Part | Execution |
|---|---|---|
| 1 | EPLZON-Board | 38.1 × 50.8 mm; A–J, 17 rows; connected five-pin groups as in the reference image |
| 2 | Jack strip | 1×8, 2.54 mm; top, openings up; C1–C8 and G1–G8 |
| 2 | Pin strip on the SuperMini | 1×8, 2.54 mm; long contacts down into the sockets |
| 1 | Connector | 2×4 open contacts, 2,54 mm; G12/H12 bis G15/H15 |
| 1 | Power Jumper | 1×2 pin header plus jumper; B11/B12 |
| 11 | Insulated wire pieces | Short thin jumper wires, e.g. AWG30; approx. 0.6 m stock |
| 1 | R1 · CS-Pull-up | 10 kΩ, axial, ≥ 1/8 W, 1–5 %; Body max. 6.5 × 2.5 mm; E10/F10 |
| 1 | SuperMini and SOP8 cable | Available; check the approved pin layout and cable assignment |

According to the EPLZON reference image, A–E and F–J are connected within each row; different rows are isolated. E–F spans 7.62 mm; C–G gives 15.24 mm between the sockets. Before soldering, verify these dimensions and the SuperMini pin spacing on the actual parts. Leave 5 V unconnected.

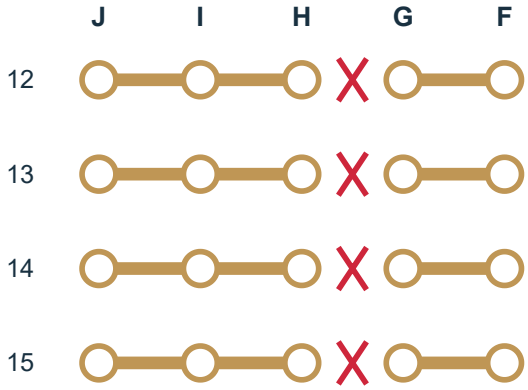
ALTERNATIVE C · PREPARE COPPER

# Four connections to separate, then bestücken

The two connector-pins of a row would otherwise be shorted. Between **G12/H12, G13/H13, G14/H14 and G15/H15** remove only the connecting copper trace. Keep the solder pads intact.

## C · FOUR SEPARATION POINTS · SOLDER SIDE

Bottom view: J–I–H | G–F. The hole labels remain those of the top side.




Between H and G: Remove the connection.

Keep the solder pads and plated-through holes intact.

**Measure before mounting:**

F ↔ G connected; H ↔ I ↔ J connected.

G ↔ H in each line separately.

Check both board sides.

If there is parallel copper there:

also separate on the other side.

Four electrical separation points; depending on copper layout, both sides may need to be worked on.

Zoom from below: The columns appear in reversed order. Line 12 remains on top. The drawing shows the required electrical state after processing.

> **Look at both sides first.** Break visible copper bridges with an appropriate fine tool without sawing through the board. If the opposite side also connects the same contacts, break those apart too. Remove copper remnants. Only populate after successful insulation measurement.

### Check the unpopulated board.

- □ In each of the four processed lines: F–G connected; H–I–J connected; G–H without through-hole.
- □ Between different lines as well as between A–E and F–J there is no additional connection.
- □ B11 and B12 are separated. They belong to two lines; the jumper must not sit across within a group of five.
- □ Do not build this coordinate layout unchanged if copper groups or hole spacing differ.

### Mount components

1. **Align without power:** Mount the SuperMini pin strips with long ends facing down. Insert the sockets at C1–C8 and G1–G8 and align them straight using the power-off module. Solder one end contact, pull the module away, and solder the remaining contacts.
2. **Connector:** Insert from the top at G12/H12 to G15/H15 and solder from below. Mark pin 1 at G12. Do not solder across the separators.
3. **Jumper:** Insert from the top at B11/B12 and solder from below. B11 is input, B12 is output; remove the jumper first.
4. **R1:** 10 kΩ between E10 (CS) and F10 (target power), body top over the middle gap. Solder bottom; no polarity. W10 and W11 connect these groups to connector 1 and 2 respectively. Do not add a second pull-up if a 10 kΩ pull-up already exists in the cable.

Choose jack height so that all contacts are securely engaged and the ESP32 bottom side, solder points, and USB connector remain free. Existing matching pin strips can be used. Plan additional time for breakouts and measurements; 15 minutes are not a guaranteed first-time setup time.

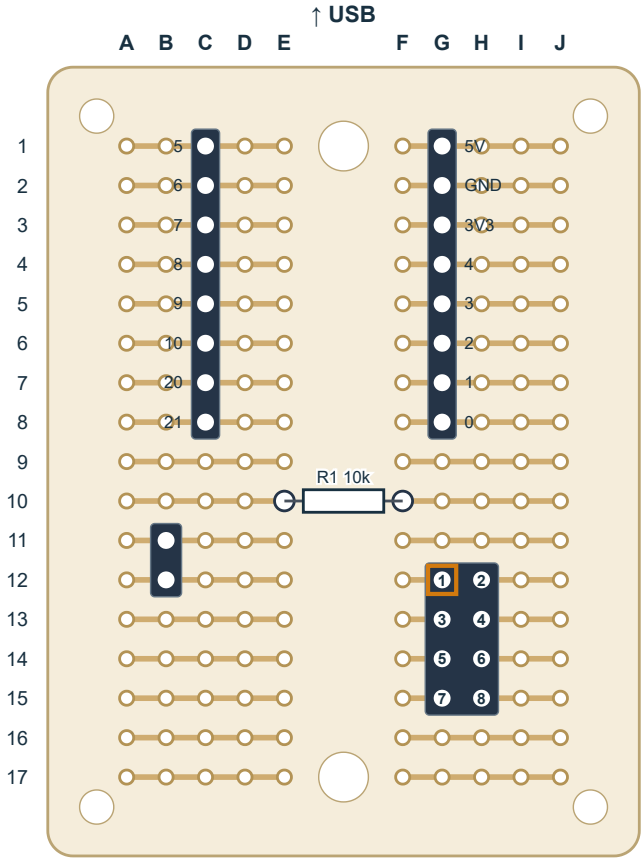
ALTERNATIVE C · COMPONENT MOUNTING & WIRE LIST

# Hole coordinates are always named from the top.

Here is the ESP32 removed. The black rows are the upward-facing sockets. Golden lines show the existing copper groups; the isolated wires are shown from below on the next side.

## C · COMPONENT SIDE · ESP32 REMOVED

All coordinates retain their designation from the top side. USB edge remains on top.




38.1 × 50.8 mm · 2.54 mm pitch

### Eleven wires · only on the solder side

W1 I4 ↔ F12 CS
W2 A1 ↔ F13 MISO
W3 A2 ↔ I14 CLK
W4 A3 ↔ I15 MOSI
W5 J2 ↔ F15 GND
W6 J3 ↔ A11 3V3
W7 E12 ↔ I12 VTARGET
W8 J12 ↔ J13 VTARGET
W9 I13 ↔ F14 VTARGET
W10 A10 ↔ G12 CS
W11 J10 ↔ H12 VTARGET

Gold: existing copper connection.
Red cross: completely separate the copper.
Color line: insulated wire below.
Only solder at both end points.
Crossing wires remain isolated.
J1 here = column J, row 1.

Alternative C · Rev. 0.4 · Unverified Build Candidate · Components on top, wires on bottom

Connector left: 1 CS, 3 MISO, 5 WP, 7 GND. Right: 2 VCC, 4 HOLD, 6 CLK, 8 MOSI. WP, HOLD and VCC get 3.3 V behind the jumper.

| Wire | Signal | Hole from | Hole after | Result |
|---|---|---|---|---|
| W1 | CS | I4 | F12 | GPIO4 → connector pin 1 |
| W2 | MISO | A1 | F13 | GPIO5 → connector pin 3 |
| W3 | CLK | A2 | I14 | GPIO6 → Jumper 6 |
| W4 | MOSI | A3 | I15 | GPIO7 → Jumper 8 |
| W5 | GND | J2 | F15 | GND → connector pin 7 |
| W6 | 3V3 | J3 | A11 | 3.3 V → Jumper Input |
| W7 | VTARGET | E12 | I12 | Jumper output → pin 2 |
| W8 | VTARGET | J12 | J13 | Connector 2 → Connector 4 (HOLD) |
| W9 | VTARGET | I13 | F14 | Connector 4 → Connector 5 (WP) |
| W10 | CS | A10 | G12 | CS → R1 at E10 |
| W11 | VTARGET | J10 | H12 | Target power → R1 to F10 |

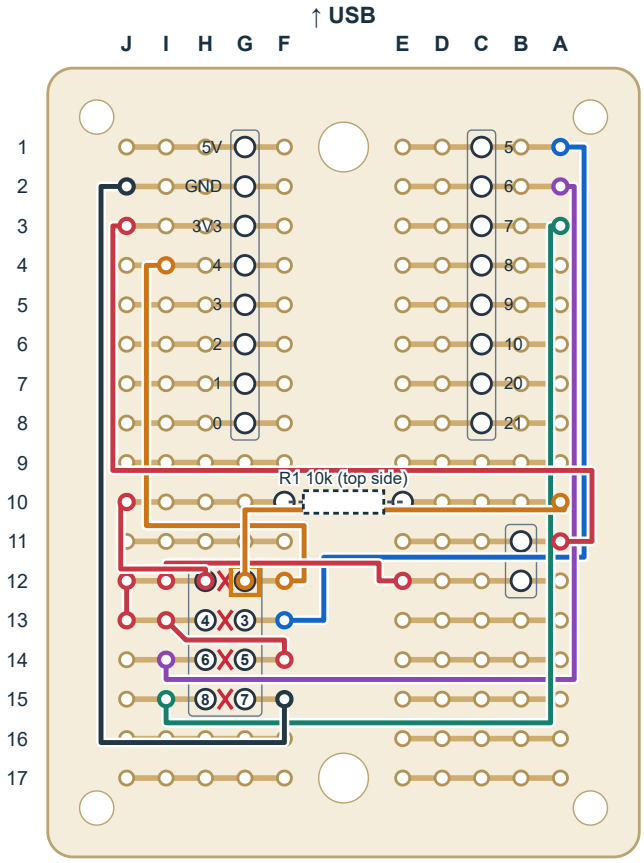
**Hole J1 means column J, row 1.** Connector pins are explicitly referred to as "Connector 1" etc. W1–W9 use free copper pads. W10/W11 are soldered together with the connector pins to G12/H12. E10/F10 connects only R1, never a wire or solder bridge. The SOP8 mapping remains 1→1, 2→8, 3→2, 4→7, 5→3, 6→6, 7→4, 8→5 (Connector→Chip).

ALTERNATIVE C · SOLDER & CHECK

# Bottom side: mirror, do not renumber

## C · SOLDER SIDE · LEFT/RIGHT MIRRORED

All coordinates retain their designation from the top side. USB edge remains on top.




38.1 × 50.8 mm · 2.54 mm pitch

### Eleven wires · only on the solder side

- W1 I4 ↔ F12 CS
- W2 A1 ↔ F13 MISO
- W3 A2 ↔ I14 CLK
- W4 A3 ↔ I15 MOSI
- W5 J2 ↔ F15 GND
- W6 J3 ↔ A11 3V3
- W7 E12 ↔ I12 VTARGET
- W8 J12 ↔ J13 VTARGET
- W9 I13 ↔ F14 VTARGET
- W10 A10 ↔ G12 CS
- W11 J10 ↔ H12 VTARGET

Gold: existing copper connection.
Red cross: completely separate the copper.
Color line: insulated wire below.
Only solder at both end points.
Crossing wires remain isolated.
J1 here = column J, row 1.

Alternative C · Rev. 0.4 · Unverified Build Candidate · Components on top, wires on bottom

Hold the USB edge on top. Fully insulated wires should be led and only soldered at the two labeled ends. Crossings are not connections. Keep free wire ends and component leads short.

## After soldering

1. **Without ESP32 and without target:** first without cable G12 → H12 on **9,5–10,5 kΩ** Measure; CS is connected via R1 to Target-VCC. Other networks must not be shorted. Then plug in the free cable and measure signal paths up to the terminal. Connectors 2/4/5 must be connected. When the jumper is open, these are separated from the 3.3-V connector; when closed, they are connected. 5 V and unused GPIOs remain without additional connections.
2. **Mechanics and power supply:** Insert the SuperMini without power; check the fit, cable direction, USB access, and jumper access. Measure 3.3 V and jumper function first without load, then under load. Keep the target battery and target USB disconnected.
3. **WebSerial:** gemeinsame Firmware and browser instructions use: serprog, SPI2, GPIO4/5/6/7, USB Serial/JTAG, WLAN off. Select the serial port via HTTPS or localhost in Chrome/Edge.
4. **Only read:** Read two complete 16-MiB dumps (each 16.777.216 bytes) at 500 kHz and compare byte by byte or via SHA-256 with each other and with a known correct dump of the same unmodified flash. No write or erase tests.

> **Jumper = power supply.** It does not disconnect the SPI lines or prevent backfeeding through them. Disconnect USB before changing the wiring. Mechanical fit, the actual copper pattern, power under load, and readback comparison still require hardware testing.

---

**Alternative C · Rev. 0.4 · Unverified Build Candidate. Drawings, CSV lists and layout-eplzon.json describe the same build.**

COMMON · WEBSERIAL & FIRST ACCESS

# WebSerial: The Armware must match the wiring

The ESP32-C3 configuration of the corresponding **TB2-Dungeon-Basis**. Overworld opens the ESP32-C3 selection and leads to this guide. An arbitrary SuperMini program will not provide a serprog connection.

> **Wire according to GPIO numbers.** Generic SuperMini pinout diagrams use different SPI function names. This guide explicitly uses: GPIO4 = CS, GPIO5 = MISO, GPIO6 = CLK, GPIO7 = MOSI.

**For all three variants: Browser → WebSerial → serial USB port of ESP32-C3 → serprog → SPI-Flash.** USB Serial/JTAG refers to the USB interface in the firmware; browser access is via WebSerial.

## One-time preparation of the SuperMini

1. Remove the target completely. esp32-serprog, Snapshot 0ca3c7e, download. The archive contains source code. ESP-IDF 6.0 or newer according to Espressif Instruction set up; use the ESP-IDF terminal.
2. In the fresh source folder `src/serprog` Run: `idf.py set-target esp32c3`.
3. In `main/main.c` the initial SPI configuration block of `#ifdef CONFIG_IDF_TARGET_ESP32` until the related `#endif` Replace entirely before the Wi-Fi section:

```
#define EEPROM_HOST SPI2_HOST
#define PIN_NUM_MISO 5
#define PIN_NUM_MOSI 7
#define PIN_NUM_CLK  6
#define PIN_NUM_CS   4
```

The unmodified source uses in the generic branch GPIO40/41/42, which do not exist on the C3.

4. `idf.py menuconfig`: Unter **Component config** → **ESP-STDIO** → **Channel for console output** the **USB Serial/JTAG Controller** Select. Deactivate secondary console, leave WLAN deactivated. GPIO18/19 remain free for USB. No additional UART cable.
5. With `idf.py build` Build and with `idf.py -p PORT flash` load onto the SuperMini; `PORT` replace it with its actual COM port. This overwrites the previous module firmware. If necessary, hold BOOT, press RESET briefly, and release BOOT; then restart normally. The target remains removed.

## First access to Overworld / Dungeon

1. After successful wiring and voltage testing, connect the adapter. Use Chrome or Edge via HTTPS or localhost; other serial programs close.
2. Select ESP32-C3, **921600 Baud** and initially **500 kHz SPI** set up. Connect and select the serial SuperMini port. With native USB, the baud rate does not determine the USB data rate. Do not use Flipper-WinUSB driver installation.
3. Only proceed if a plausible W25Q128 is detected. Read completely twice and save both files. Each must **16.777.216 Byte** be large. Compare the SHA-256 of both files, additionally with a known correct dump of the same unmodified chip. **Do not delete or write.**

```
Get-Item .\read-1.bin, .\read-2.bin | Select-Object Name, Length
Get-FileHash .\read-1.bin, .\read-2.bin, .\reference.bin -Algorithm SHA256
```

Adjust filenames to the stored files. Equality of two new dumps alone does not exclude a systematic read error. In case of discrepancies, first check the wiring, power supply and clock.

## Test bench and sources

**Statically checked:** Connector-to-chip mapping, firmware pins, mirroring, wiring grid, and jumper separation. **Open:** specific module matching, solder points, voltage under load and real read comparisons. This guide contains no pre-built firmware image. Status: **unverified candidate**.

- Winbond: external CS-pull-up, typically 10 kΩ.
- Project verification: TB2-Dev-Board `docs/hardware.md` and `Physical-pin-coordinates.csv`; TB2 Dungeon `Website/assets/cards/interface/template.js`; Overworld guide. As of 10.10.2026.
- Nologo: SuperMini component side with USB on top (5V/GND/3V3 right; do not use the underside pinout as a top view) Espressif: ESP32-C3 GPIO · serprog source position.

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**Rev. 0.4 · Schematics and instructions for the minimal perfboard adapter. The SVG files are editable; the HTML file contains all schematics and styles.**